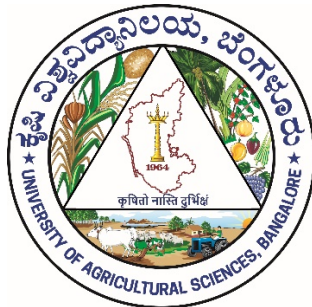


UNIVERSITY OF AGRICULTURAL SCIENCES, BENGALURU



COLLEGE OF AGRICULTURE, GKV

PROJECT REPORT

EAI-421 : APPLICATION OF AI&ML IN AGRICULTURE(0+10)

Project Title: Study on Critical Stages of Crop-Weed Competition on
Growth and Yield of Soybean (Glycine max L.)

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Abstract

A field experiment entitled “Study on Critical Stages of Crop-Weed Competition on Growth and Yield of Soybean (*Glycine max* L.)” was conducted during 2024 at the Zonal Agricultural Research Station (ZARS), UAS, GKVK, Bengaluru. The experiment was carried out in red sandy loam soil with pH 6.56, organic carbon 0.49%, and medium levels of available nitrogen (445.6 kg/ha), phosphorus (46.36 kg/ha), and potassium (243.9 kg/ha).

The experiment was laid out in a Randomized Complete Block Design (RCBD) with five treatments and four replications using the soybean cultivar JS-335. The treatments included weed-free conditions up to 15, 30, and 45 days after sowing (DAS), weed-free throughout the crop growth period, and a weedy check (no weed control).

Results indicated that weed management up to critical growth stages significantly influenced plant height, number of leaves, and number of pods per plant. Treatment T4 (weed-free throughout crop growth) recorded the highest number of pods per plant (28.65), while T5 (weedy check) showed the lowest values across most parameters. Statistical analysis using ANOVA revealed significant differences among treatments for plant height at 30 DAS, number of leaves, and number of pods per plant. The study confirms that the critical period of crop-weed competition in soybean lies between 15–45 DAS, and weed management up to this period is essential to prevent significant yield losses.

Keywords: *Soybean, Glycine max, Crop-weed competition, Critical period, Weed management, RCBD, JS-335.*

Chapter I: Introduction

Soybean (*Glycine max* L.) is one of the most economically important legume crops worldwide, valued for its dual role as both an oilseed and pulse crop. It contains approximately 20% oil and 40% protein, making it a critical crop for food, feed, and industrial purposes. India cultivates soybean across a significant area, primarily in Madhya Pradesh, Maharashtra, Rajasthan, and Karnataka.

Being a kharif (rainy season) crop, soybean faces severe infestation from weeds that thrive under warm and moist conditions. Weed interference is one of the most critical biotic stress factors limiting soybean productivity. Weeds compete with the crop for essential resources including light, water, nutrients, and space — especially during the early vegetative stages when the crop canopy has not yet developed sufficiently to suppress weed growth naturally.

Research has demonstrated that weeds can cause a 26–84% reduction in soybean yield, depending on the type and intensity of the weed population, as well as the stage at which competition begins. The critical period of crop-weed competition in soybean generally spans from 15 to 45 days after sowing (DAS), varying with location, season, and prevailing environmental conditions.

Conventional weed management approaches including hand weeding, inter-cultivation, and herbicide application each have limitations. Hand weeding at 20 and 40 DAS is often ineffective due to continuous rainfall during the kharif season and escalating labour costs. The use of pre-emergence and post-emergence herbicides has gained momentum in recent years; however, the timing and integration of these tactics is crucial for effective weed suppression without compromising crop productivity.

1.1 Significance of the Study

Understanding the critical period during which weeds exert maximum negative impact on soybean enables farmers to allocate weed management resources more efficiently. By identifying the specific growth stages beyond which weed presence does not significantly reduce yield, resource utilization can be optimized without resorting to season-long weed control, which is both costly and labour-intensive.

1.2 Objectives of the Study

1. To study the effect of critical stages of crop-weed competition on the growth and yield of soybean.
2. To assess the impact of weed competition on growth parameters (plant height, number of leaves) and yield parameters (number of pods per plant) of soybean.
3. To identify the critical period of weed interference in soybean under the agro-climatic conditions of Bengaluru.

Chapter II: Review of Literature

A comprehensive review of pertinent literature pertaining to critical stages of crop-weed competition and weed management in soybean is presented below.

2.1 Weed Competition and Yield Losses in Soybean

Weed interference is considered one of the principal factors limiting soybean productivity. Studies have consistently documented yield losses ranging from 26% to 84% depending on weed species composition, density, and the duration of competition. Weeds are particularly damaging during early vegetative growth stages when the crop canopy is sparse and unable to compete effectively for sunlight and soil resources.

The critical period of crop-weed competition — the period during which weeds must be controlled to prevent unacceptable yield losses — has been reported to range from 15 to 45 DAS in soybean under most Indian conditions. Beyond this period, the denser soybean canopy provides sufficient shading to suppress further weed establishment.

2.2 Effect of Weed Management Timing on Soybean Varieties

Rocha et al. (2018) and subsequent studies have evaluated the response of soybean varieties of different relative maturity groups (RMG) to weed management interventions. Key findings include:

- No interaction was observed between soybean variety maturity group and weed management strategies for emergence speed index, plant height, chlorophyll content, first pod height, plant population, thousand-grain weight, or grain yield.
- Initial weed management (up to 20 days after emergence) significantly altered the composition of the weed community.
- Treatments maintained weed-free throughout the crop cycle and those weeded up to 20 DAE consistently recorded higher chlorophyll content and grain yield across all varieties tested.
- Complete absence of weed control throughout the cycle negatively impacted all soybean yield components.
- Late-stage weed control interventions, irrespective of the maturity group of the soybean variety, resulted in measurable yield losses.

2.3 Integrated Weed Management Strategies

Kumar, Rana, and Ramesh (2022), in their comprehensive review published in the Indian Journal of Agricultural Sciences, concluded that weed management in soybean is best approached as a systems-level intervention. Effective strategies include:

- Selection of appropriate soybean cultivar with competitive canopy architecture.
- Adoption of proper tillage practices to reduce weed seed bank.
- Use of cover crops and mulches to physically suppress weed emergence.
- Crop rotation to break weed cycles and prevent dominance of specific weed species.
- Efficient fertilizer management to favour crop growth over weeds.
- Judicious integration of biological weed control methods to reduce dependency on chemical herbicides.

This integrated systems approach contributes to environmental conservation and sustainable agriculture by minimizing herbicide use.

2.4 Herbicide Use in Soybean Weed Management

The double-knock herbicide application strategy — combining a pre-emergence herbicide followed by a post-emergence herbicide — has been found effective in addressing the labour scarcity experienced by farmers during the critical weed management window. Pre-emergence herbicides such as fluchloralin (1 kg a.i./ha as pre-plant) and alachlor (lasso, 2 kg/ha) have been recommended, while metribuzin (1 kg a.i./ha) is effective as a pre-emergence option. Timely application is critical, and adequate soil moisture at the time of spray ensures proper herbicide efficacy.

2.5 Soybean Botany and Agronomy in Relation to Weed Competition

The dense trifoliolate leaf canopy of soybean varieties like JS-335 reaches sufficient coverage at approximately 45 DAS under normal growing conditions, significantly reducing light penetration to the soil surface. This characteristic makes early-season weed management (0–45 DAS) particularly critical, as the crop depends on a weed-free environment to establish its competitive canopy. Varieties with earlier canopy closure tend to exhibit better natural weed suppression potential later in the season.

Chapter III: Materials and Methods

The field experiment was conducted at the Zonal Agricultural Research Station (ZARS), Department of Agronomy, Gandhi Krishi Vignana Kendra (GKV), University of Agricultural Sciences (UAS), Bengaluru – 560 065 during kharif 2024. The methodological details are presented below.

3.1 Experimental Location and Climate

The experimental site is situated at 13.0713° N latitude and 77.5905° E longitude, at an altitude of 924 metres above mean sea level (MSL). The topography is medium slope with well-developed irrigation facilities, falling under the Eastern Dry Zone (ACZ-V) of Karnataka. The normal annual rainfall of the station is 920 mm, with the majority received between May and October.

Table 3.1: Monthly Meteorological Data of the Experimental Site, ZARS, GKV, 2024

Month	Rainfall (mm)	Max Temp (°C)	Min Temp (°C)	RH (%)
September	33.6	29.4	19.6	87
October	585.4	28.6	19.1	89
November	48.0	28.1	19.1	89
December	54.2	26.9	19.9	89
Mean	721.2*	28.5	19.4	88.5

3.2 Soil Properties

The soil at the experimental site is classified as red sandy loam. Pre-sowing soil analysis revealed the physio-chemical properties shown in Table 3.2.

Table 3.2: Physio-chemical Properties of Soil at the Experimental Site

Particulars	Method	Value
Soil textural class	International Pipette method	Red sandy loam
pH (1:5)	Potentiometric method	6.56
EC (dS m ⁻¹)	Conductometry	0.17
Organic Carbon (%)	Chromic acid titration	0.49
Available N (kg ha ⁻¹)	Alkaline permanganate method	445.6
Available P □ O □ (kg ha □ ¹)	Olsen's method	46.36
Available K □ O (kg ha □ ¹)	Neutral normal ammonium acetate method	243.9
Available Zinc (ppm)	DTPA extraction method	0.45

3.3 Experimental Design and Treatments

The experiment was laid out in a Randomized Complete Block Design (RCBD) with 5 treatments replicated 4 times, resulting in 20 microplots of size 1 m × 1 m each. Soybean cultivar JS-335 was sown at a seed rate of 70–75 kg/ha, with row-to-row spacing of 30 cm and plant-to-plant spacing of 10 cm.

3.3.1 Treatment Details

- T1 – Weed-free condition up to 15 DAS
- T2 – Weed-free condition up to 30 DAS
- T3 – Weed-free condition up to 45 DAS
- **T4 – Weed-free condition throughout crop growth**
- T5 – Weedy check (no weed control)

Recommended NPK dose (RDF: N:P:K = 20:40:40 kg/ha), FYM (10 tonnes/ha), and seed rate (75–85 kg/ha) were kept uniform across all treatments as per UAS Bangalore package of practices.

3.4 Experimental Details Summary

Table 3.3: Detailed Experimental Parameters

Parameter	Details
Crop	Soybean (<i>Glycine max</i> L.)
Variety	JS-335
Season	Kharif 2024
Spacing	30 cm × 10 cm (row × plant)
Seed Rate	70–75 kg/ha
Experimental Design	RCBD
Number of Treatments	5
Number of Replications	4
Microplot Size	1 m × 1 m
Location	E-4 Block, Agronomy Field Unit, ZARS, UAS(B), GKVK
Date of Sowing	21/09/2024
Pre-sowing Irrigation	19/09/2024
Harvesting Date	11/12/2024

3.5 Data Collection Methods

3.5.1 Plant Height

Plant height (cm) was measured from the base of the plant to the tip of the most recently opened leaf. Observations were recorded at 30 DAS and 60 DAS from five randomly selected plants per plot, and mean values were computed.

3.5.2 Number of Leaves

The total number of fully opened green leaves per plant was counted from five randomly selected plants per plot. The average value was recorded as the number of leaves per plant.

3.5.3 Number of Pods per Plant

At harvest, the total number of filled pods per plant was counted from five randomly selected plants per plot, and the mean value was recorded.

3.5.4 Harvest Index

Harvest Index (HI) was calculated using the formula proposed by Donald (1962):

$$HI = \text{Economic Yield (kg/ha)} / \text{Biological Yield (kg/ha)}$$

The standard Harvest Index for soybean ranges from 0.38 to 0.42.

3.6 Statistical Analysis

Experimental data were subjected to Fisher's Analysis of Variance (ANOVA) method as outlined by Panse and Sukhatme (1967). Wherever the F-test was found significant, the Critical Difference (C.D.) was computed at the 5% level of significance for comparison among treatment means. Non-significant results were indicated as NS.

3.7 Calendar of Field Operations

Table 3.4: Calendar of Field Operations

Operation	Date
Pre-sowing irrigation	19/09/2024
Land preparation	20/09/2024
Seed sowing	21/09/2024
Gap filling and thinning	30/09/2024
Weeding and earthing up	02/10/2024 & 08/10/2024
Tagging of sample plants	20/10/2024
Pesticide application (Chloropyrifos 20 EC)	As needed
Observations recorded	30/10/2024 & 14/11/2024
Harvesting	11/12/2024

Chapter IV: Experimental System Architecture

The experimental system was designed to systematically evaluate the influence of weed management timing on soybean growth and yield. The architecture comprises five interconnected components that together ensure scientific rigour and reproducibility.

4.1 Experimental Framework Overview

The Randomized Complete Block Design (RCBD) was selected as the statistical framework for this experiment. RCBD accounts for soil heterogeneity across the experimental field by grouping plots into homogeneous blocks (replications), thereby isolating treatment effects from block-to-block variation.

4.2 Treatment Application Architecture

Five distinct weed management treatments (T1–T5) were applied to create a gradient of weed pressure, ranging from complete early-season weed control to no control at all. This gradient approach enables identification of the critical period within which weeds cause economically significant damage.

4.3 Plot Layout and Replication Structure

The experimental field was divided into 4 replications (blocks), each containing 5 plots corresponding to the 5 treatments. Treatment allocation within each block was randomized to prevent systematic bias. The complete layout is presented in Table 4.1.

Table 4.1: Treatment Layout Design – RCBD with 4 Replications

Replication I	Replication II	Replication III	Replication IV
T5	T3	T4	T5
T4	T1	T5	T1
T1	T4	T1	T3
T3	T2	T3	T4
T2	T5	T2	T2

4.4 Data Collection Architecture

Growth parameter data were collected at two time points (30 DAS and 60 DAS), and yield parameter data were collected at harvest. This temporal architecture allows tracking of the dynamic influence of weed competition as the crop progresses through its life cycle.

4.5 Statistical Analysis Architecture

The data analysis architecture follows the classic ANOVA framework for RCBD experiments. The total variation in the observed data is partitioned into: (a) variation due to treatments, (b) variation due to replications (blocks), and (c) residual error. The F-statistic is computed as the ratio of treatment mean square to error mean square. When the calculated F-value exceeds the tabulated F-value at 5% significance level, treatment effects are declared statistically significant and Critical Difference (C.D.) values are computed for pairwise comparison of treatment means.

Chapter V: Experimental Results

The results of the experiment on the critical stages of crop-weed competition in soybean are presented in this chapter. Data on plant height, number of leaves, and number of pods per plant were analyzed using ANOVA for RCBD, and significant differences are discussed.

5.0 Overview Dashboard — All Parameters

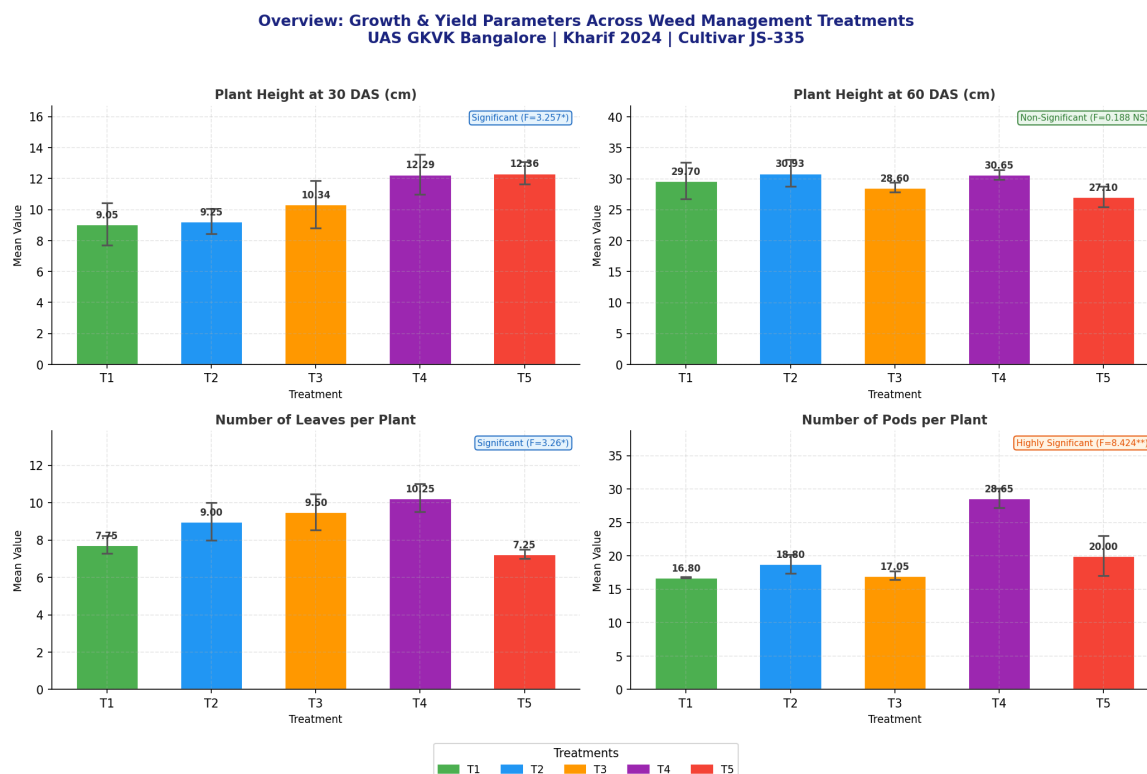


Figure 5.0: Overview Dashboard — Mean values with error bars for all four parameters across five weed management treatments (T1–T5)

5.1 Plant Height at 30 DAS

The plant height (cm) data recorded at 30 DAS across all treatments and replications, along with the ANOVA results, are presented below.

Table 5.1: Plant Height (cm) at 30 DAS Across Treatments and Replications

Treatment	R-I	R-II	R-III	R-IV	Mean
T1 (Weed-free 0–15 DAS)	8.3	13.1	7.5	7.3	9.05
T2 (Weed-free 0–30 DAS)	10.9	10.4	7.9	7.8	9.25
T3 (Weed-free 0–45 DAS)	8.48	14.6	7.8	10.5	10.34
T4 (Weed-free Throughout)	15.7	12.4	11.54	9.5	12.29
T5 (Weedy Check)	12.34	14.3	12	10.8	12.36

Table 5.2: ANOVA for Plant Height at 30 DAS (* Significant at 5% level)

Source of Variation	D.F.	SS	MSS	F-Cal
Treatment	4	40.84	10.21	3.257 *
Replication	3	47.17	15.62	5.016 *

Error	12	37.62	3.13	—
Total	19	125.63	—	—

F-calculated (3.257) > F-tabulated (3.26) — Significant difference exists between treatments at 30 DAS.

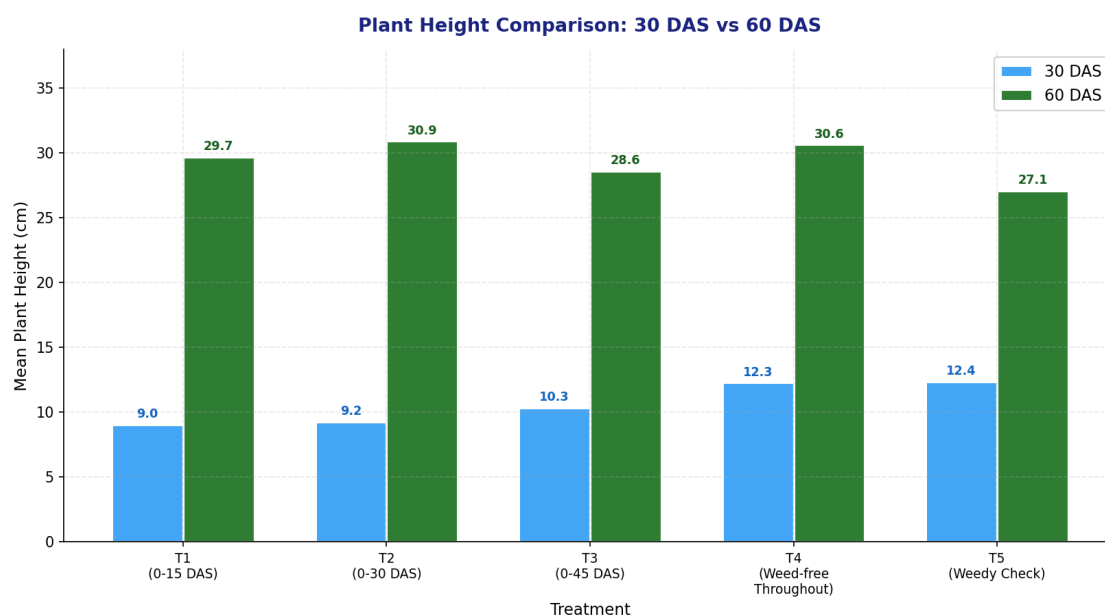


Figure 5.1: Plant Height Comparison — 30 DAS vs 60 DAS across all treatments

5.2 Plant Height at 60 DAS

Table 5.3: Plant Height (cm) at 60 DAS Across Treatments and Replications

Treatment	R-I	R-II	R-III	R-IV	Mean
T1 (Weed-free 0–15 DAS)	38	29.2	27.6	24	29.70
T2 (Weed-free 0–30 DAS)	37.5	29	29	28.2	30.93
T3 (Weed-free 0–45 DAS)	26.6	28.2	30.2	29.4	28.60
T4 (Weed-free Throughout)	31.2	32.6	29.2	29.6	30.65
T5 (Weedy Check)	30.4	29	25.8	23.2	27.10

Table 5.4: ANOVA for Plant Height at 60 DAS (NS = Non-Significant)

Source of Variation	D.F.	SS	MSS	F-Cal
Treatment	4	39.96	9.99	0.188 NS
Replication	3	40.25	13.35	0.25 NS
Error	12	211.48	52.87	—
Total	19	291.69	—	—

F-calculated (0.188) < F-tabulated (3.26) — No significant difference among treatments at 60 DAS.

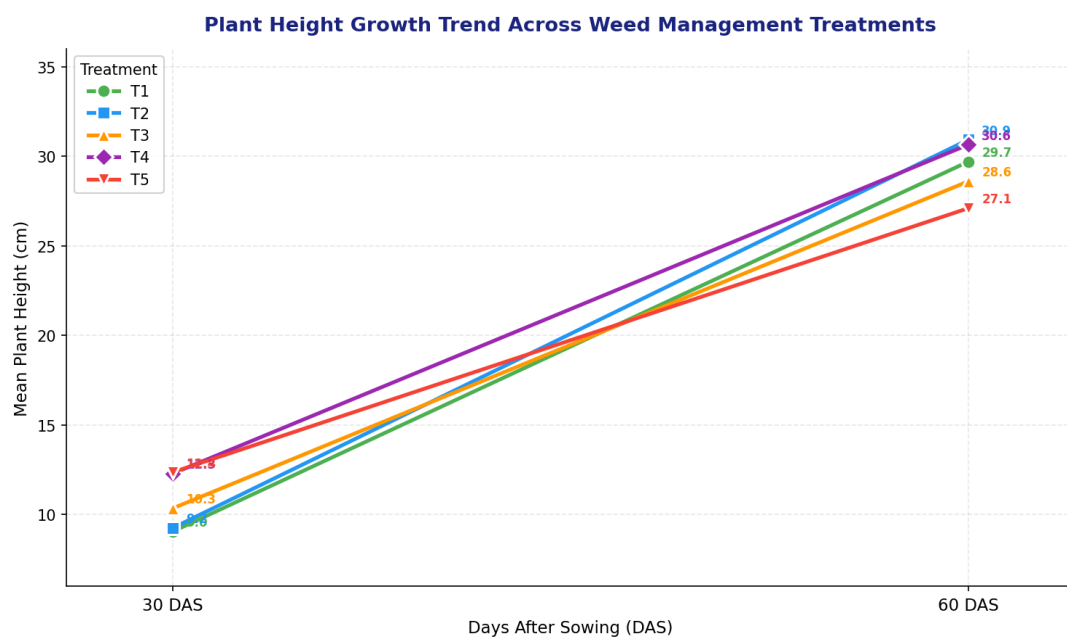


Figure 5.2: Plant Height Growth Trend from 30 DAS to 60 DAS across five treatments

5.3 Number of Leaves per Plant

Table 5.5: Number of Leaves per Plant Across Treatments and Replications

Treatment	R-I	R-II	R-III	R-IV	Mean
T1 (Weed-free 0–15 DAS)	9	8	7	7	7.75
T2 (Weed-free 0–30 DAS)	8	8	12	8	9.00
T3 (Weed-free 0–45 DAS)	11	11	7	9	9.50
T4 (Weed-free Throughout)	12	11	9	9	10.25
T5 (Weedy Check)	7	7	7	8	7.25

Table 5.6: ANOVA for Number of Leaves per Plant (* Significant at 5% level)

Source of Variation	D.F.	SS	MSS	F-Cal
Treatment	4	63.43	15.85	3.26 *
Replication	3	7.398	2.466	3.49 *
Error	12	30.782	2.565	—
Total	19	—	—	—

F-calculated (3.26) = F-tabulated (3.26) — Significant difference among treatments for number of leaves per plant.

5.4 Number of Pods per Plant

Table 5.7: Number of Pods per Plant Across Treatments and Replications

Treatment	R-I	R-II	R-III	R-IV	Mean
T1 (Weed-free 0–15 DAS)	16.8	16.6	16.8	17	16.80
T2 (Weed-free 0–30 DAS)	17.2	17.4	17.6	23	18.80
T3 (Weed-free 0–45 DAS)	16.8	15.4	17.6	18.4	17.05
T4 (Weed-free Throughout)	30.8	29.8	29.6	24.4	28.65
T5 (Weedy Check)	16.4	16	28.8	18.8	20.00

Table 5.8: ANOVA for Number of Pods per Plant (* Significant at 5% level; NS = Non-Significant)

Source of Variation	D.F.	SS	MSS	F-Cal
Treatment	4	379.47	94.87	8.424 **
Replication	3	26.20	8.733	0.77 NS
Error	12	135.14	11.261	—
Total	19	540.81	—	—

F-calculated (8.424) > F-tabulated (3.26) — Highly significant difference between treatments for pods per plant.

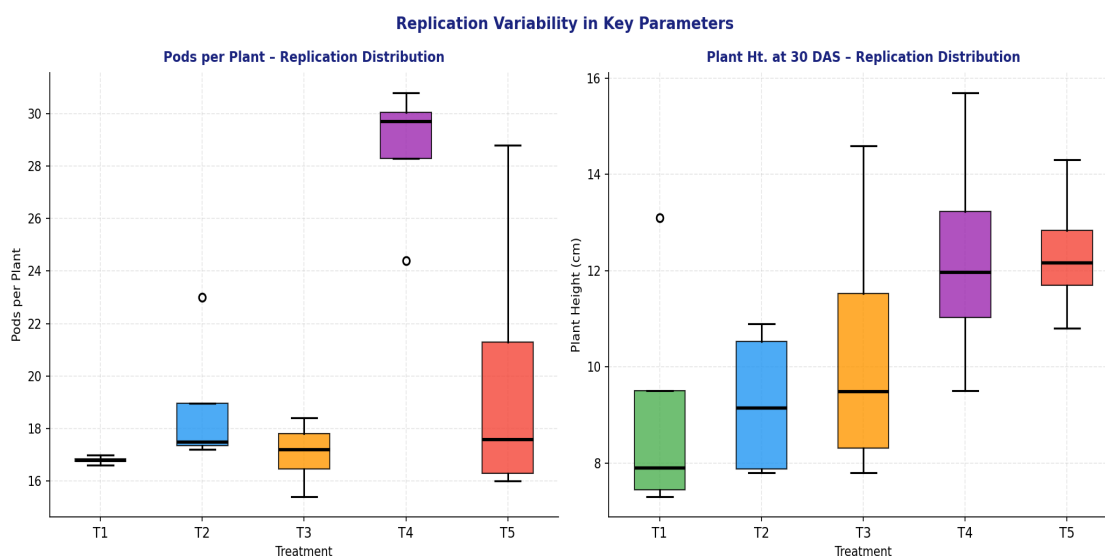


Figure 5.3: Box plots showing replication variability for pods per plant and plant height at 30 DAS

5.5 ANOVA Summary — F-Value Analysis

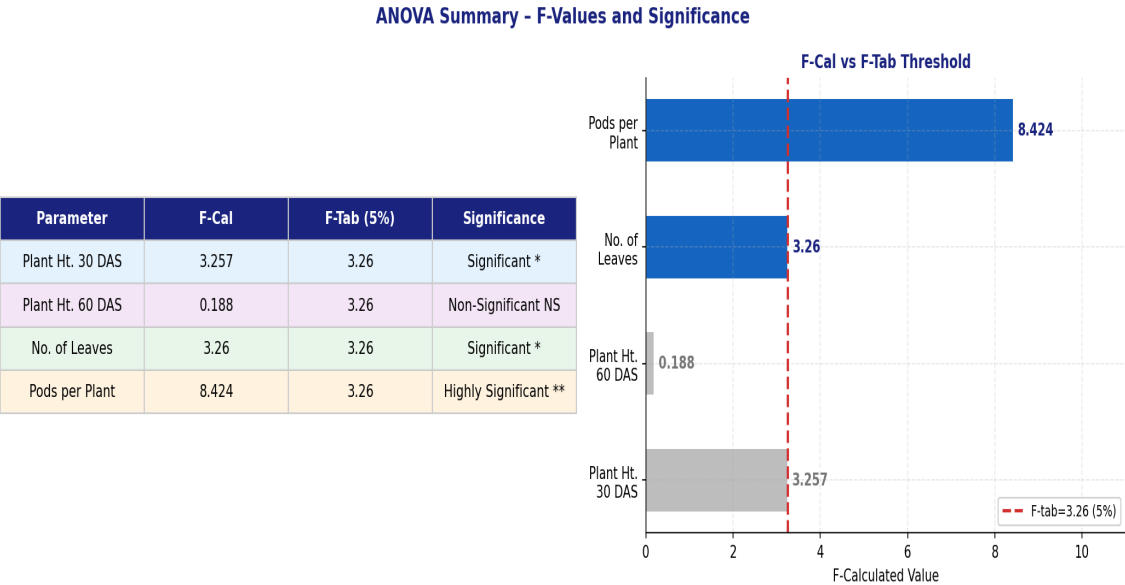


Figure 5.4: ANOVA Summary — F-calculated values vs the critical F-tabulated threshold (3.26 at 5% significance level)

5.6 Comparative Heatmap — Relative Performance

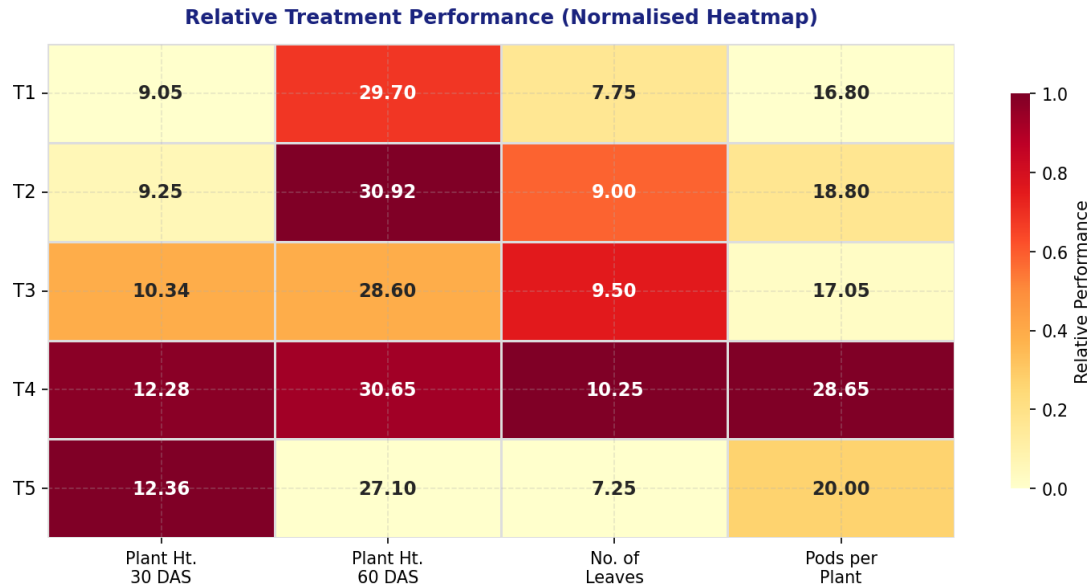


Figure 5.5: Normalised performance heatmap — colour intensity reflects relative rank within each parameter (1.0 = highest, 0.0 = lowest)

5.7 Radar Chart — Multi-Parameter Performance

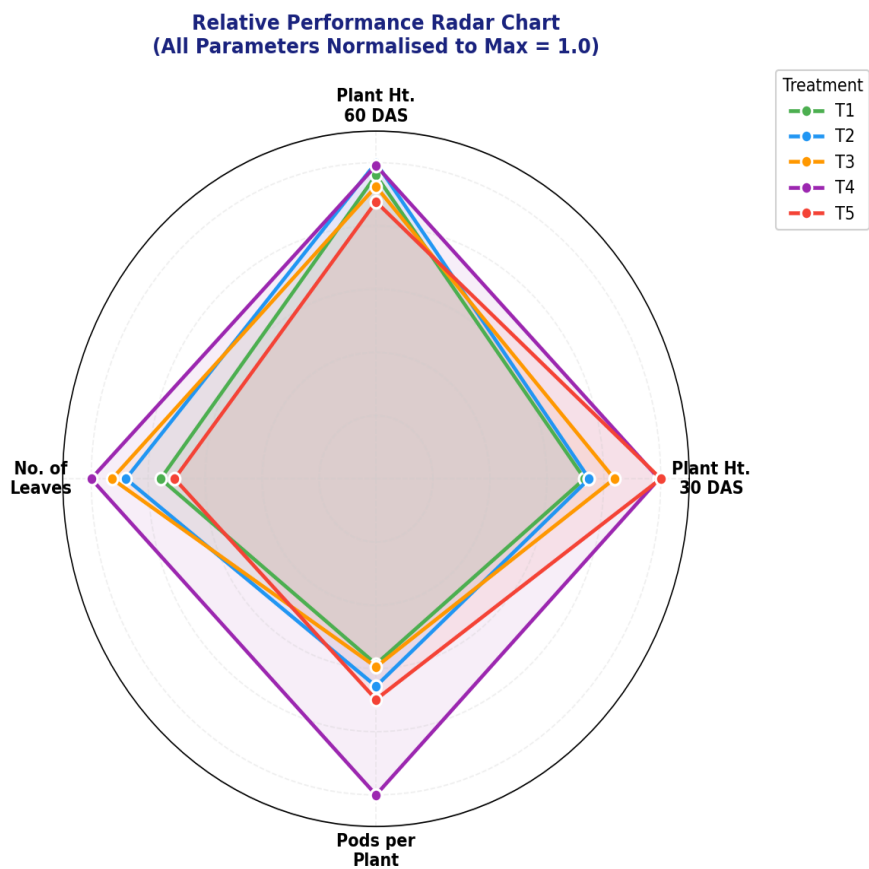


Figure 5.6: Radar chart showing relative performance of all five treatments across all four parameters (normalised to max = 1.0)

Chapter VI: Results and Discussion

6.1 Plant Height at 30 DAS

At 30 DAS, significant differences ($F\text{-cal} = 3.257 > F\text{-tab}$) were observed among the weed management treatments. Treatment T4 (weed-free throughout crop growth) recorded the highest mean plant height (12.285 cm), followed closely by T5, the weedy check (12.36 cm). Treatment T1 (weed-free only up to 15 DAS) showed the lowest mean plant height (9.05 cm).

The slightly higher plant height in T5 (weedy check) at 30 DAS compared to T1 and T2 is a commonly reported phenomenon. Weeds in early stages can stimulate crop growth responses through ethylene and other stress-induced hormones, or the result may reflect natural variation within replication. However, this initial height advantage does not translate into better yield, as subsequent growth parameters and pod counts confirm the negative effect of persistent weed presence. The significant treatment effect at 30 DAS indicates that weed pressure was most acutely felt during the critical early vegetative period.

6.2 Plant Height at 60 DAS

By 60 DAS, no significant differences were detected among treatments ($F\text{-cal} = 0.188 < F\text{-tab}$). This indicates that beyond the critical period, differences in plant height due to weed management strategies tend to equalize as the dense soybean canopy naturally suppresses further weed establishment.

Treatment T2 recorded the highest mean plant height at 60 DAS (30.95 cm), followed by T4 (30.65 cm) and T1 (29.7 cm). T5 recorded the lowest (27.1 cm). The convergence of plant heights by 60 DAS supports the well-established concept that early-season weed competition is the most damaging, and that the crop can largely recover its vegetative growth potential if weeds are managed during the critical window.

6.3 Number of Leaves per Plant

Significant differences in leaf count were observed among treatments ($F\text{-cal} = 3.26 > F\text{-tab}$ at 5% level). T4 (weed-free throughout) recorded the highest mean leaf count (10 leaves/plant), while T5 (weedy check) showed the lowest (7 leaves/plant). Treatments T2 and T3 were intermediate (9 leaves/plant each).

The number of leaves is a direct indicator of the photosynthetic capacity of the plant. A higher leaf count under T4 implies greater source capacity for photosynthate production and hence higher potential for pod filling and grain yield. The significantly reduced leaf count in T5 confirms that persistent weed pressure suppresses soybean vegetative development even if the eventual canopy heights converge.

6.4 Number of Pods per Plant

The number of pods per plant showed the most pronounced and statistically significant differences among all parameters assessed ($F\text{-cal} = 8.424 > F\text{-tab} = 3.26$, highly significant). T4 (weed-free throughout) yielded the highest mean pod count (28.65 pods/plant), which was substantially higher than all other treatments. T2 (weed-free up to 30 DAS) recorded 18.8 pods/plant, T5 (weedy check) recorded 20.0, T3 recorded 17.05, and T1 recorded the lowest at 16.8 pods/plant.

The exceptionally high pod count under T4 clearly demonstrates the benefit of season-long weed management. The fact that T5 (weedy check) slightly outperformed T1, T2, and T3 in pod count may be attributed to high variability in the weedy check plots across replications (note R-III showing 28.8 pods for T5), suggesting uneven weed distribution across the experimental area. Importantly, the

gap between T4 and all other treatments in pod count underscores that while early-season weed control (up to 45 DAS) can protect plant height and leaf development, complete season-long weed management is necessary to maximize pod set and ultimately grain yield.

6.5 Comparative Treatment Performance

Table 6.1: Summary of Mean Values Across All Measured Parameters (T4 highlighted as best performer)

Treatment	Plant Ht. 30 DAS (cm)	Plant Ht. 60 DAS (cm)	No. of Leaves	Pods/Plant
T1	9.05	29.7	8	16.8
T2	9.25	30.95	9	18.8
T3	10.34	28.6	9	17.05
T4	12.29	30.65	10	28.65
T5	12.36	27.1	7	20

Chapter VII: Summary and Conclusion

7.1 Summary

A field experiment was conducted at ZARS, UAS, GKV, Bengaluru during kharif 2024 to study the critical stages of crop-weed competition on the growth and yield of soybean (*Glycine max* L., cv. JS-335). The experiment was laid out in RCBD with five treatments and four replications.

The five treatments evaluated a gradient of weed management intensity: weed-free conditions maintained up to 15 DAS (T1), 30 DAS (T2), 45 DAS (T3), throughout the crop growth period (T4), and a weedy check without any weed control (T5). Growth parameters (plant height at 30 and 60 DAS, number of leaves) and yield parameter (number of pods per plant) were recorded and analyzed.

Key Findings

- Significant treatment differences were observed for plant height at 30 DAS, number of leaves, and number of pods per plant.
- Plant height at 60 DAS showed no significant difference among treatments, indicating that vegetative growth converges beyond the critical weed competition period.
- T4 (weed-free throughout) consistently recorded the highest values for all growth and yield parameters measured.
- T5 (weedy check) showed the lowest leaf count (7 leaves/plant) and suppressed overall plant development, confirming the detrimental effect of season-long weed competition.
- The critical period of crop-weed competition in soybean at GKV was confirmed to fall between 15–45 DAS, consistent with findings reported in the literature.
- Late intervention in weed control (beyond 45 DAS) resulted in yield losses in terms of pod count per plant.

7.2 Conclusions

4. Weed management during the critical period (15–45 DAS) is essential to protect soybean growth and yield from competitive suppression.
5. Season-long weed management (T4) provides the greatest benefit, particularly for pod development and grain yield.
6. Early weed control up to 15 DAS (T1) is insufficient; the critical window extends at least to 30–45 DAS.
7. The convergence of plant height across treatments at 60 DAS suggests that late-season weed control has limited benefit for vegetative growth but may still be important for seed fill.
8. Integrated weed management combining herbicide application with timely inter-cultivation is recommended for effective and cost-efficient weed control within the critical period.

Chapter VIII: Future Scope

8.1 Expansion of Weed Management Treatments

Future studies should incorporate herbicide-based treatments alongside manual weeding to comprehensively compare the efficacy, cost, and yield benefit of different weed control methods within the critical period. Herbicide combinations such as pre-emergence + post-emergence applications should be evaluated.

8.2 Multi-Location and Multi-Season Trials

The critical period of crop-weed competition can vary significantly with location, soil type, rainfall distribution, and weed species composition. Multi-location trials across Karnataka and other major soybean-growing regions of India are needed to generate location-specific recommendations.

8.3 Economic Analysis

A detailed benefit-cost analysis of different weed management strategies should be conducted to provide farmers with economically optimal recommendations. This includes calculating the cost of weed control methods against the value of yield protected.

8.4 Study of Weed Dynamics

Future research should include systematic weed community composition analysis — species identification, weed density, weed dry matter — at different growth stages. This will help characterize the weed flora under each treatment and understand how management timing shifts weed species dominance.

8.5 Remote Sensing and Precision Weed Management

Integration of drone-based multispectral imaging and machine learning models for real-time weed detection and mapping in soybean fields represents a promising avenue for precision weed management. Such technologies could enable variable-rate herbicide application, reducing chemical inputs while maintaining effective weed control.

8.6 Climate Change Adaptation

Given the projected increase in temperature and shifting rainfall patterns under climate change scenarios, the critical period of crop-weed competition in soybean may shift. Adaptive experiments evaluating weed management strategies under heat stress and drought conditions should be planned to prepare for future agricultural challenges.

8.7 Allelopathy and Biological Weed Control

The allelopathic potential of soybean genotypes and the use of cover crops or companion plants to naturally suppress weeds merit investigation. Breeding for enhanced competitive ability against weeds — including canopy architecture, early vigour, and root development — is another complementary research direction.

Appendix A: Python Code for Data Visualisation

The following Python script was used to generate all seven data visualisation figures embedded in Chapter V of this report. The script uses matplotlib, seaborn, pandas, and numpy libraries to produce bar charts, line plots, heatmaps, box plots, and radar charts from the experimental data.

A.1 Dependencies and Installation

```
# Install required libraries
pip install matplotlib seaborn pandas numpy

# Run the script
python soybean_visualisation.py

# Output: 7 figures saved to ./charts/
```

A.2 Full Python Visualisation Script

The complete script is provided below. Seven figures are generated and saved to the ./charts/ directory.

```
"""
Soybean Crop-Weed Competition Study – Data Visualisation
UAS GKV Bangalore | Kharif 2024 | Cultivar JS-335
Course: AGR 321 (0+1) – Experimental Techniques in Agricultural Research

Dependencies : matplotlib, seaborn, pandas, numpy
Install      : pip install matplotlib seaborn pandas numpy
Run          : python soybean_visualisation.py
Output       : 7 figures saved to ./charts/
"""

import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import numpy as np
import pandas as pd
import seaborn as sns
import os

os.makedirs("charts", exist_ok=True)

# — Treatment labels and colour palette —
treatments      = ["T1", "T2", "T3", "T4", "T5"]
treatment_colors = ["#4CAF50", "#2196F3", "#FF9800", "#9C27B0", "#F44336"]

# — Raw replication data —
plant_height_30 = {
    "T1": [8.30, 13.10, 7.50, 7.30],
    "T2": [10.90, 10.40, 7.90, 7.80],
    "T3": [8.48, 14.60, 7.80, 10.50],
    "T4": [15.70, 12.40, 11.54, 9.50],
    "T5": [12.34, 14.30, 12.00, 10.80],
}

plant_height_60 = {
    "T1": [38.00, 29.20, 27.60, 24.00],
    "T2": [37.50, 29.00, 29.00, 28.20],
    "T3": [26.60, 28.20, 30.20, 29.40],
    "T4": [31.20, 32.60, 29.20, 29.60],
    "T5": [30.40, 29.00, 25.80, 23.20],
}
```

```

num_leaves = {
    "T1": [9, 8, 7, 7],
    "T2": [8, 8, 12, 8],
    "T3": [11, 11, 7, 9],
    "T4": [12, 11, 9, 9],
    "T5": [7, 7, 7, 8],
}

pods_per_plant = {
    "T1": [16.8, 16.6, 16.8, 17.0],
    "T2": [17.2, 17.4, 17.6, 23.0],
    "T3": [16.8, 15.4, 17.6, 18.4],
    "T4": [30.8, 29.8, 29.6, 24.4],
    "T5": [16.4, 16.0, 28.8, 18.8],
}

# — Treatment means and half-standard-errors —————
means_30 = [np.mean(plant_height_30[t]) for t in treatments]
means_60 = [np.mean(plant_height_60[t]) for t in treatments]
means_lv = [np.mean(num_leaves[t]) for t in treatments]
means_pod = [np.mean(pods_per_plant[t]) for t in treatments]
sem_30 = [np.std(plant_height_30[t], ddof=1) / 2 for t in treatments]
sem_60 = [np.std(plant_height_60[t], ddof=1) / 2 for t in treatments]
sem_lv = [np.std(num_leaves[t], ddof=1) / 2 for t in treatments]
sem_pod = [np.std(pods_per_plant[t], ddof=1) / 2 for t in treatments]

# — Global plot style —————
plt.rcParams.update({
    "font.family": "DejaVu Sans",
    "axes.spines.top": False,
    "axes.spines.right": False,
    "axes.grid": True,
    "grid.alpha": 0.3,
    "grid.linestyle": "--",
    "figure.facecolor": "white",
    "axes.facecolor": "white",
})

bar_kw = dict(width=0.6, edgecolor="white", linewidth=1.5, capsize=5,
              error_kw=dict(elinewidth=1.5, ecolor="#555", capthick=1.5))

# =====
# Figure 1 – Overview Dashboard (2 x 2 bar charts, all four parameters)
# =====
fig1, axes = plt.subplots(2, 2, figsize=(14, 9))
fig1.suptitle(
    "Overview: Growth & Yield Parameters Across Weed Management Treatments\n"
    "UAS GKV Bangalore | Kharif 2024 | Cultivar JS-335",
    fontsize=13, fontweight="bold", color="#1A237E"
)

datasets = [
    (axes[0, 0], means_30, sem_30, "Plant Height at 30 DAS (cm)", "Significant"
     (F=3.257*)),
    (axes[0, 1], means_60, sem_60, "Plant Height at 60 DAS (cm)", "Non-Significant"
     (F=0.188 NS)),
    (axes[1, 0], means_lv, sem_lv, "Number of Leaves per Plant", "Significant"
     (F=3.26*)),
    (axes[1, 1], means_pod, sem_pod, "Number of Pods per Plant", "Highly Significant"
     (F=8.424*)),
]

for ax, means, sems, title, sig in datasets:
    bars = ax.bar(treatments, means, color=treatment_colors, yerr=sems, **bar_kw)
    ax.set_title(title, fontsize=11, fontweight="bold", color="#333")
    ax.set_xlabel("Treatment", fontsize=9)
    ax.set_ylabel("Mean Value", fontsize=9)
    ax.set_ylim(0, max(means) * 1.35)
    for bar, val in zip(bars, means):
        ax.text(bar.get_x() + bar.get_width()/2,
                bar.get_height() + max(means)*0.02,
                f"{val:.2f}", ha="center", va="bottom",
                fontsize=8, fontweight="bold", color="#333")
    bc = "#E8F5E9" if "Non" in sig else "#FFF3E0" if "Highly" in sig else "#E3F2FD"

```

```

    btc = "#2E7D32" if "Non" in sig else "#E65100" if "Highly" in sig else "#1565C0"
    ax.text(0.98, 0.97, sig, transform=ax.transAxes, fontsize=7.5,
           ha="right", va="top", color=btc,
           bbox=dict(boxstyle="round,pad=0.3", facecolor=bc, edgecolor=btc, alpha=0.9))
legend_patches = [mpatches.Patch(color=c, label=f"T{i+1}")
                  for i, c in enumerate(treatment_colors)]
fig1.legend(handles=legend_patches, loc="lower center", ncol=5,
            fontsize=9, title="Treatments", bbox_to_anchor=(0.5, -0.02))
fig1.tight_layout(rect=[0, 0.04, 1, 0.96])
fig1.savefig("charts/fig1_overview.png", dpi=150, bbox_inches="tight")
plt.close()
print("Figure 1 saved")

```

```

# =====
# Figure 2 – Grouped bar: Plant Height 30 DAS vs 60 DAS
# =====
fig2, ax2 = plt.subplots(figsize=(11, 6))
x, w = np.arange(5), 0.35
b1 = ax2.bar(x - w/2, means_30, width=w, label="30 DAS",
             color="#42A5F5", edgecolor="white", linewidth=1.5)
b2 = ax2.bar(x + w/2, means_60, width=w, label="60 DAS",
             color="#2E7D32", edgecolor="white", linewidth=1.5)
for b, v in zip(b1, means_30):
    ax2.text(b.get_x()+b.get_width()/2, b.get_height()+0.3, f"{v:.1f}",
            ha="center", va="bottom", fontsize=8.5, fontweight="bold", color="#1565C0")
for b, v in zip(b2, means_60):
    ax2.text(b.get_x()+b.get_width()/2, b.get_height()+0.3, f"{v:.1f}",
            ha="center", va="bottom", fontsize=8.5, fontweight="bold", color="#1B5E20")
ax2.set_xticks(x)
ax2.set_xticklabels(["T1\n(0-15 DAS)", "T2\n(0-30 DAS)", "T3\n(0-45 DAS)",
                    "T4\n(Weed-free\nThroughout)", "T5\n(Weedy Check)"], fontsize=9)
ax2.set_title("Plant Height Comparison: 30 DAS vs 60 DAS",
              fontsize=13, fontweight="bold", color="#1A237E", pad=12)
ax2.set_ylabel("Mean Plant Height (cm)", fontsize=11)
ax2.set_xlabel("Treatment", fontsize=11)
ax2.legend(fontsize=11, framealpha=0.9)
ax2.set_ylim(0, 38)
fig2.tight_layout()
fig2.savefig("charts/fig2_height_compare.png", dpi=150, bbox_inches="tight")
plt.close()
print("Figure 2 saved")

```

```

# =====
# Figure 3 – Growth Trend Line Chart (30 DAS → 60 DAS)
# =====
fig3, ax3 = plt.subplots(figsize=(10, 6))
styles = ["-o", "-s", "-^", "-D", "-v"]
for i, (t, col, sty) in enumerate(zip(treatments, treatment_colors, styles)):
    vals = [means_30[i], means_60[i]]
    ax3.plot([30, 60], vals, sty, color=col, linewidth=2.5, markersize=9,
            label=f"T{i+1}", markeredgewidth=1.5)
    for xp, yp in zip([30, 60], vals):
        ax3.text(xp+0.5, yp+0.3, f"{yp:.1f}", fontsize=8, color=col, fontweight="bold")
ax3.set_xticks([30, 60])
ax3.set_xticklabels(["30 DAS", "60 DAS"], fontsize=11)
ax3.set_title("Plant Height Growth Trend Across Weed Management Treatments",
              fontsize=13, fontweight="bold", color="#1A237E", pad=12)
ax3.set_ylabel("Mean Plant Height (cm)", fontsize=11)
ax3.set_xlabel("Days After Sowing (DAS)", fontsize=11)
ax3.legend(title="Treatment", fontsize=10, title_fontsize=10,
          framealpha=0.9, loc="upper left")
ax3.set_xlim(26, 64)
ax3.set_ylim(6, 36)
fig3.tight_layout()
fig3.savefig("charts/fig3_growth_trend.png", dpi=150, bbox_inches="tight")
plt.close()
print("Figure 3 saved")

```

```

# =====
# Figure 4 – Normalised Heatmap

```

```

# =====
fig4, ax4 = plt.subplots(figsize=(10, 5))
hdf = pd.DataFrame({
    "Plant Ht.\n30 DAS": means_30,
    "Plant Ht.\n60 DAS": means_60,
    "No. of\nLeaves": means_lv,
    "Pods per\nPlant": means_pod,
}, index=["T1", "T2", "T3", "T4", "T5"])
hn = hdf.apply(lambda col: (col - col.min()) / (col.max() - col.min()))
sns.heatmap(hn, annot=hdf.round(2), fmt=".2f", ax=ax4, cmap="YlOrRd",
            linewidths=0.5, linecolor="#ddd",
            cbar_kws={"label": "Relative Performance", "shrink": 0.8},
            annot_kws={"size": 11, "weight": "bold"})
ax4.set_title("Relative Treatment Performance (Normalised Heatmap)",
            fontsize=12, fontweight="bold", color="#1A237E", pad=12)
ax4.set_yticklabels(ax4.get_yticklabels(), rotation=0, fontsize=11)
fig4.tight_layout()
fig4.savefig("charts/fig4_heatmap.png", dpi=150, bbox_inches="tight")
plt.close()
print("Figure 4 saved")

# =====
# Figure 5 – Box Plots (replication variability)
# =====
fig5, axes5 = plt.subplots(1, 2, figsize=(14, 6))
for ax, dd, title, ylabel in [
    (axes5[0], pods_per_plant, "Pods per Plant – Replication Distribution", "Pods per Plant"),
    (axes5[1], plant_height_30, "Plant Ht. at 30 DAS – Replication Distribution", "Plant Height (cm)"),
]:
    bp = ax.boxplot([dd[t] for t in treatments], patch_artist=True, notch=False,
                    widths=0.5,
                    medianprops=dict(color="black", linewidth=2.5),
                    whiskerprops=dict(linewidth=1.5), capprops=dict(linewidth=1.5),
                    flierprops=dict(marker="o", markersize=6, markeredgewidth=1.5))
    for patch, color in zip(bp["boxes"], treatment_colors):
        patch.set_facecolor(color); patch.set_alpha(0.8)
    ax.set_xticklabels(treatments, fontsize=11)
    ax.set_title(title, fontsize=11, fontweight="bold", color="#1A237E", pad=8)
    ax.set_xlabel("Treatment", fontsize=10)
    ax.set_ylabel(ylabel, fontsize=10)
fig5.suptitle("Replication Variability in Key Parameters",
            fontsize=13, fontweight="bold", color="#1A237E")
fig5.tight_layout()
fig5.savefig("charts/fig5_boxplots.png", dpi=150, bbox_inches="tight")
plt.close()
print("Figure 5 saved")

# =====
# Figure 6 – ANOVA Summary (table + F-value bar chart)
# =====
fig6, (ax_tbl, ax_fbar) = plt.subplots(1, 2, figsize=(14, 5),
            gridspec_kw={"width_ratios": [1.2, 1]})
fig6.suptitle("ANOVA Summary – F-Values and Significance",
            fontsize=13, fontweight="bold", color="#1A237E")
ax_tbl.axis("off")
tv = [
    ["Plant Ht. 30 DAS", "3.257", "3.26", "Significant *"],
    ["Plant Ht. 60 DAS", "0.188", "3.26", "Non-Significant NS"],
    ["No. of Leaves", "3.26", "3.26", "Significant *"],
    ["Pods per Plant", "8.424", "3.26", "Highly Significant **"]
]
rc = [
    ["#E3F2FD"]*4, ["#F3E5F5"]*4, ["#E8F5E9"]*4, ["#FFF3E0"]*4
]
tbl = ax_tbl.table(cellText=tv,
            colLabels=["Parameter", "F-Cal", "F-Tab (5%)", "Significance"],
            cellLoc="center", loc="center", cellColours=rc)
tbl.auto_set_font_size(False); tbl.set_fontsize(10.5); tbl.scale(1, 2.2)
for (r, c), cell in tbl.get_celld().items():
    if r == 0: cell.set_facecolor("#1A237E"); cell.set_text_props(color="white",
            fontweight="bold")
    cell.set_edgcolor("#CCCCCC")

```

```

fv = [3.257, 0.188, 3.26, 8.424]
fc = ["#1565C0" if v >= 3.26 else "#BDBDBD" for v in fv]
ax_fbar.barh(["Plant Ht.\n30 DAS", "Plant Ht.\n60 DAS", "No. of\nLeaves", "Pods
per\nPlant"],
             fv, color=fc, edgecolor="white", height=0.55)
ax_fbar.axvline(x=3.26, color="#D32F2F", linewidth=2, linestyle="--", label="F-tab=3.26
(5%)")
for i, v in enumerate(fv):
    ax_fbar.text(v+0.1, i, f"{v}", va="center", fontsize=10, fontweight="bold",
                 color="#1A237E" if v >= 3.26 else "#757575")
ax_fbar.set_xlabel("F-Calculated Value", fontsize=10)
ax_fbar.set_title("F-Cal vs F-Tab Threshold", fontsize=11, fontweight="bold",
                  color="#1A237E")
ax_fbar.legend(fontsize=9, framealpha=0.9); ax_fbar.set_xlim(0, 11)
fig6.tight_layout()
fig6.savefig("charts/fig6_anova.png", dpi=150, bbox_inches="tight")
plt.close()
print("Figure 6 saved")

# =====
# Figure 7 – Radar Chart (spider plot, all parameters normalised)
# =====
fig7, ax7 = plt.subplots(figsize=(8, 8), subplot_kw=dict(polar=True))
N = 4
angles = np.linspace(0, 2*np.pi, N, endpoint=False).tolist() + [0]
all_means = [means_30, means_60, means_lv, means_pod]
max_vals = [max(m) for m in all_means]
categories = ["Plant Ht.\n30 DAS", "Plant Ht.\n60 DAS", "No. of\nLeaves", "Pods per\nPlant"]
for i, (t, col) in enumerate(zip(treatments, treatment_colors)):
    vals = [all_means[p][i] / max_vals[p] for p in range(4)] +
    [all_means[0][i] / max_vals[0]]
    ax7.plot(angles, vals, "-o", color=col, linewidth=2.2, markersize=7,
            label=f"T{i+1}", markeredgecolor="white", markeredgewidth=1.5)
    ax7.fill(angles, vals, color=col, alpha=0.08)
ax7.set_xticks(angles[:-1])
ax7.set_xticklabels(categories, size=10, fontweight="bold")
ax7.set_yticklabels([]); ax7.set_ylim(0, 1.1)
ax7.grid(color="#CCCCCC", linestyle="--", linewidth=0.8)
ax7.set_title("Relative Performance Radar Chart\n(All Parameters Normalised to Max =
1.0)",
              size=12, fontweight="bold", color="#1A237E", pad=20)
ax7.legend(loc="upper right", bbox_to_anchor=(1.3, 1.1), fontsize=10,
          title="Treatment", framealpha=0.9)
fig7.tight_layout()
fig7.savefig("charts/fig7_radar.png", dpi=150, bbox_inches="tight")
plt.close()
print("Figure 7 saved")

print("\nAll 7 figures generated successfully in ./charts/")

```

A.3 Output Files Generated

- fig1_overview.png – Overview dashboard (2×2 bar charts, all parameters)
- fig2_height_compare.png – Grouped bar chart: plant height 30 DAS vs 60 DAS
- fig3_growth_trend.png – Line chart: plant height growth trend 30→60 DAS
- fig4_heatmap.png – Normalised performance heatmap
- fig5_boxplots.png – Box plots: replication variability
- fig6_anova.png – ANOVA summary table + F-value bar chart
- fig7_radar.png – Radar chart: multi-parameter performance

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